

Network Administration Lab Manual

Durham College – Network Administration Lab

Baseline Setup

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Part 1 – Generate

Step 1: Configure VMware Network Settings

VMware Network Editor Configuration for VMnet8

The image shows the VMware Network Editor interface for configuring VMnet8. The main window displays a table of network settings and configuration options. Below the main window, two sub-windows are open: 'NAT Settings' and 'Domain Name Server (DNS)'.

Virtual Network Editor

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Bridged	Auto-bridging	-	-	-
VMnet1	Host-only	-	Connected	Enabled	192.168.229.0
VMnet8	NAT	NAT	Connected	Enabled	10.1.1.0

Buttons: Add Network..., Remove Network, Rename Network...

VMnet Information

☐ Bridged (connect VMs directly to the external network)
Bridged to: Automatic Automatic Settings...

☒ NAT (shared host's IP address with VMs) NAT Settings...

☐ Host-only (connect VMs internally in a private network)

☒ Connect a host virtual adapter to this network
Host virtual adapter name: VMware Network Adapter VMnet8

☒ Use local DHCP service to distribute IP address to VMs DHCP Settings...

Subnet IP: 10 . 1 . 1 . 0 Subnet mask: 255 . 255 . 255 . 0

Buttons: Restore Defaults, Import..., Export..., OK, Cancel, Apply, Help

NAT Settings

Network: vmnet8
Subnet IP: 10.1.1.0
Subnet mask: 255.255.255.0
Gateway IP: 10 . 1 . 1 . 2

Port Forwarding

Host Port	Type	Virtual Machine IP Address	Description
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Buttons: Add..., Remove, Properties

Advanced

☒ Allow active FTP
☒ Allow any Organizationally Unique Identifier

UDP timeout (in seconds): 30

Config port: 0

☐ Enable IPv6
IPv6 prefix: fd15:4ba5:5a2b:1008::/64

Buttons: DNS Settings..., NetBIOS Settings..., OK, Cancel, Help

Domain Name Server (DNS)

☐ Auto detect available DNS servers

Policy: Order

Timeout (sec): 2

Retries: 3

DNS Servers

Preferred DNS server: 127 . 0 . 0 . 1

Alternate DNS server 1: . . .

Alternate DNS server 2: . . .

Buttons: OK, Cancel, Help

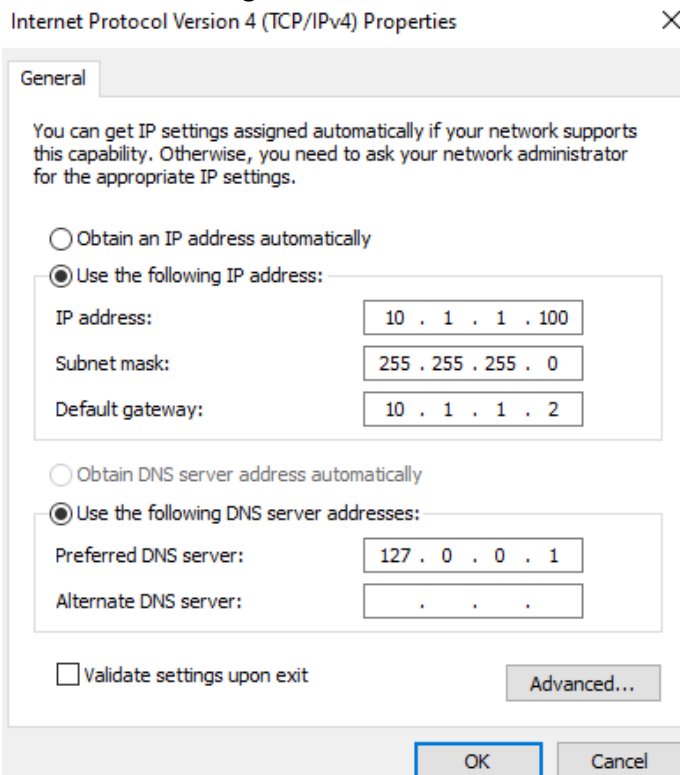
Step 2: Create the Virtual Machines

VMware VM Summary showing ServerDC network as VMnet8 (NAT)



Step 3: Configure Windows Server 2022 (ServerDC)

ServerDC IPv4 Configuration Window



Step 4: Configure Windows Server 2019 (Server01)

Server01 IP Configuration Window

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 10 . 1 . 1 . 101

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 10 . 1 . 1 . 2

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 10 . 1 . 1 . 100

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

Step 5: Configure Windows 10 and 11 Clients

1. Boot both Windows 10 and Windows 11 VMs.
2. Complete installation (use local account).
3. Go to Network & Internet Settings → Ethernet → Properties.
4. Confirm IPv4 is set to Obtain IP address automatically.
5. Confirm the IPs fall within the 10.1.1.101–199 range after DHCP is active later.
6. Rename the computers as: Windows 10 → Win10Client, Windows 11 → Win11Client.
7. Restart each VM.
8. [Screenshot Placeholder: Windows 10 Network Settings showing DHCP assigned address]

Checkpoint

At this stage, you should have:

- ✓ All four VMs created
- ✓ Static IPs configured for ServerDC and Server01
- ✓ NAT network functional
- ✓ Windows 10 and 11 ready for DHCP assignment

Part 2 – Open

Step 1: Add Roles and Features on ServerDC

ServerDC – Add Roles and Features Summary

ROLES AND SERVER GROUPS			
Roles: 4 Server groups: 1 Servers total: 1			
 AD DS 1	 DHCP 1	 DNS 1	 File and Storage Services 1
 Manageability	 Manageability	 Manageability	 Manageability
Events	Events	Events	Events
Services	Services	Services	Performance
Performance	Performance	Performance	BPA results
BPA results	BPA results	BPA results	

Step 2: Promote ServerDC to a Domain Controller

Active Directory Domain Services Configuration Wizard – New Forest contoso.com

 Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
ServerDC

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

☐ Add a domain controller to an existing domain

☐ Add a new domain to an existing forest

☒ Add a new forest

Specify the domain information for this operation

Root domain name:

[More about deployment configurations](#)

< Previous

Next >

Install

Cancel

Step 3: Verify Domain Controller and DNS Configuration

DNS Manager Showing contoso.com Forward Lookup Zone

The screenshot shows the DNS Manager console with the following structure:

- DNS
 - SERVERDC
 - Forward Lookup Zones
 - contoso.com (selected)

The main pane displays the following records:

Name	Type	Data	Timestamp
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[19] serverdc.contoso.com...	static
(same as parent folder)	Name Server (NS)	serverdc.contoso.com.	static
(same as parent folder)	Host (A)	10.1.1.100	11/4/2025
serverdc	Host (A)	10.1.1.100	static

Step 4: Configure and Authorize DHCP

DHCP Manager Showing Authorized Scope "Classroom"

The screenshot shows the DHCP Manager console with the following structure:

- DHCP
 - serverdc.contoso.com
 - IPv4
 - Scope [10.1.1.0] Classroom (selected)

The main pane displays the "Contents of Scope" for the selected scope:

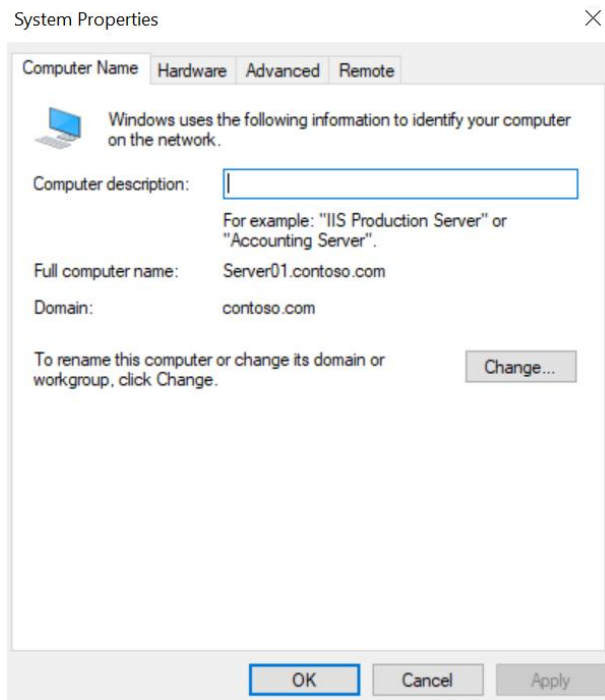
- Address Pool
- Address Leases
- Reservations
- Scope Options
- Policies

The right pane shows the "Actions" menu with the following options:

- Scope [10.1.1.0] Classroom (selected)
- More Actions

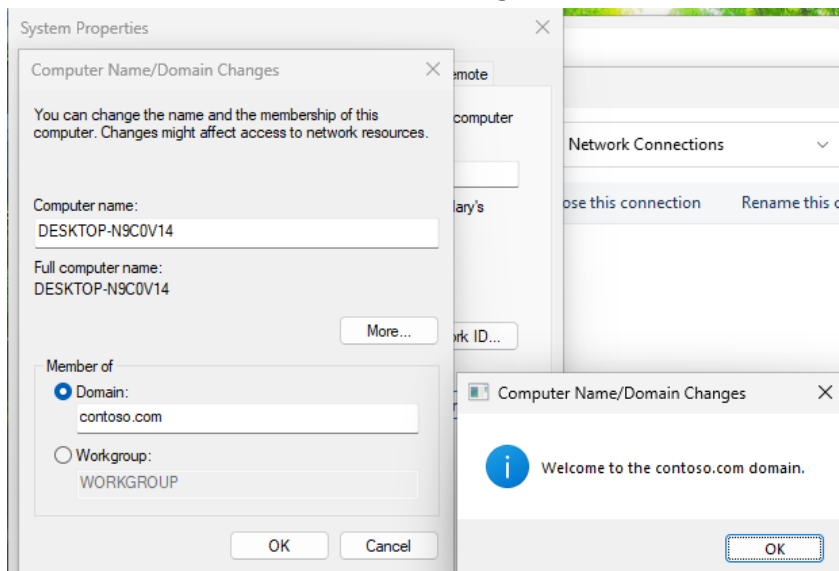
Step 5: Join Server01 to the Domain

System Properties – Computer Name/Domain Changes Confirmation



Step 6: Join Windows 10 and Windows 11 Clients to the Domain

Windows 11 Domain Join Success Message



Step 7: Confirm Network Connectivity

Successful Ping Tests From ServerDC to All Clients

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.1850]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping server01

Pinging server01.contoso.com [10.1.1.101] with 32 bytes of data:
Reply from 10.1.1.101: bytes=32 time<1ms TTL=128
Reply from 10.1.1.101: bytes=32 time<1ms TTL=128
Reply from 10.1.1.101: bytes=32 time<1ms TTL=128
Reply from 10.1.1.101: bytes=32 time<1ms TTL=128

Ping statistics for 10.1.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>ping win10client

Pinging win10client.contoso.com [10.1.1.131] with 32 bytes of data:
Reply from 10.1.1.131: bytes=32 time<1ms TTL=128
Reply from 10.1.1.131: bytes=32 time<1ms TTL=128
Reply from 10.1.1.131: bytes=32 time<1ms TTL=128
Reply from 10.1.1.131: bytes=32 time<1ms TTL=128

Ping statistics for 10.1.1.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>ping win11client

Pinging win11client.contoso.com [10.1.1.130] with 32 bytes of data:
Reply from 10.1.1.130: bytes=32 time<1ms TTL=128
Reply from 10.1.1.130: bytes=32 time<1ms TTL=128
Reply from 10.1.1.130: bytes=32 time<1ms TTL=128
Reply from 10.1.1.130: bytes=32 time<1ms TTL=128

Ping statistics for 10.1.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Checkpoint

At this stage, you should have:

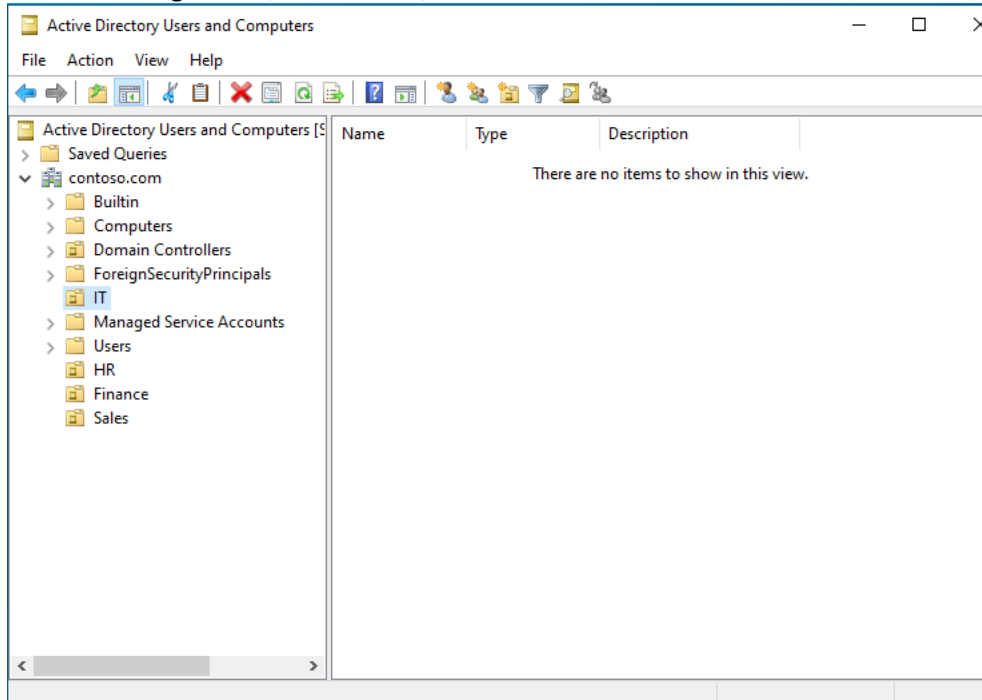
- ✓ ServerDC configured as a Domain Controller, DNS, and DHCP Server
- ✓ Server01, Win10Client, and Win11Client successfully joined to the domain
- ✓ Verified DHCP lease, DNS resolution, and domain logon across all hosts

Part 3 – Execute

Step 1: Create Organizational Units (OUs)

Method A – Using ADUC (Active Directory Users and Computers)

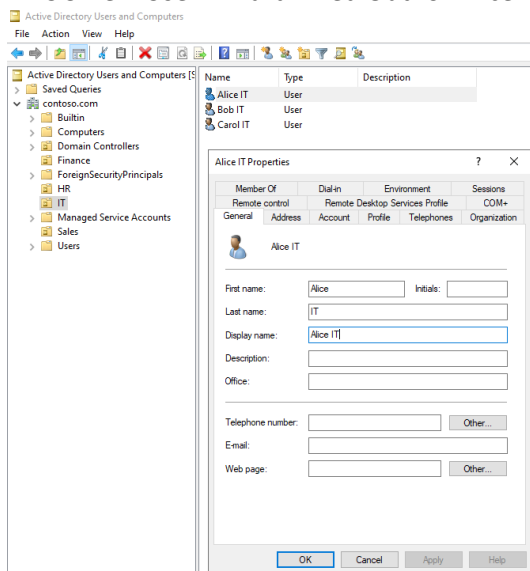
ADUC Showing OUs IT, HR, Finance, and Sales



Step 2: Create Users via GUI

Method A – Using ADUC

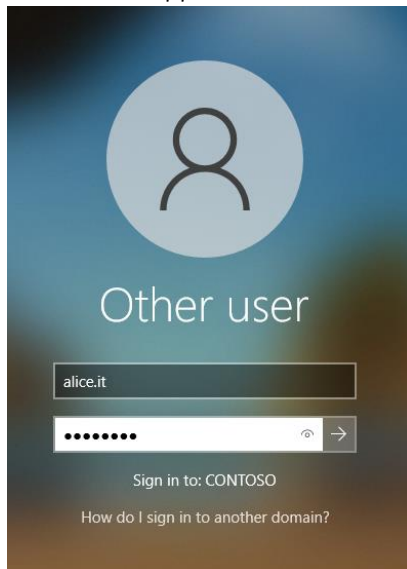
ADUC New User Wizard Filled Out for Alice IT



Step 4: Log In Using Newly Created Users

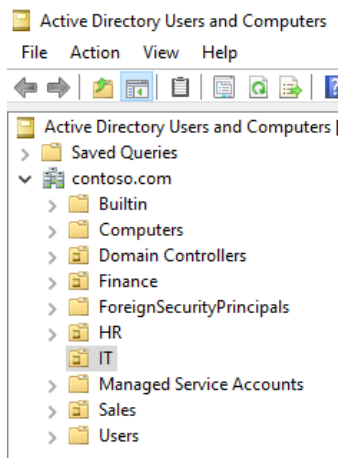
Windows 10 Login Screen with Domain User Credentials

Password: Pa\$\$w0rd

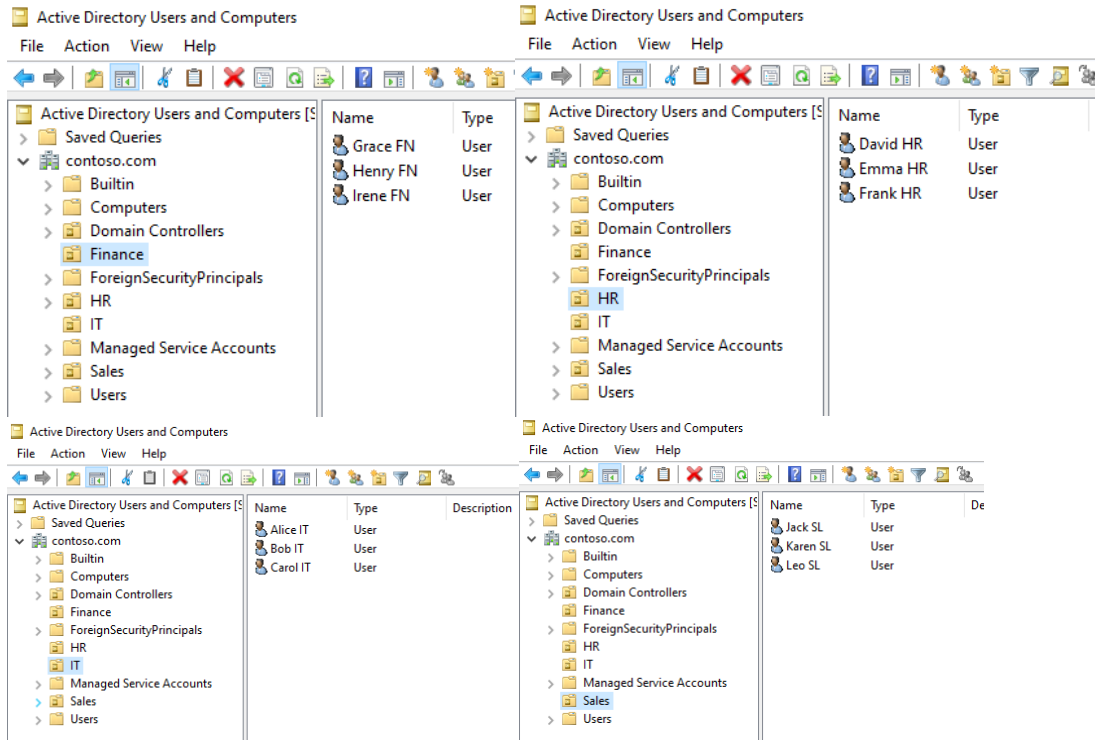


Deliverables

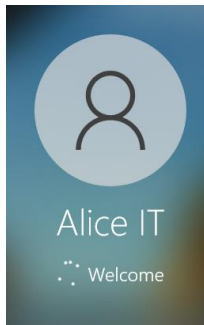
1. Screenshot showing all four OUs in ADUC or ADAC.



2. Screenshot showing three users per OU.



3. Screenshot of user login from any client PC.



4. Reflection (3–4 sentences): Explain the main differences between creating users in ADUC, ADAC, and using dsadd.exe. Which method do you think scales better in enterprise environments and why?

Active Directory Users and Computers (ADUC) is good for creating single or small numbers of users using the user interface. Active Directory Administrative Center has PowerShell integration that allows for creating single users and a large group of users. Dsadd.exe is especially useful for creating a large group of users using scripts with the legacy command-line tool. I think ADAC would scale the best in enterprise environments with its automation using PowerShell scripts.